

"Because we are human, we use hands": Understanding People's Expectations and Mental Models of Virtual Object Interaction in AR Headsets

Alejandro Delgado

Department of Computer Science and
Engineering
University of South Florida
Tampa, Florida, USA
alejandrodel@usf.edu

Hetvi Shah

Department of Computer Science and
Engineering
University of South Florida
Tampa, Florida, USA
hetvi@usf.edu

Md Mehedi Hasan Jibon

Department of Computer Science and
Engineering
University of South Florida
Tampa, Florida, USA
jibon@usf.edu

Anzhelika Kurnikova

Department of Computer Science and
Engineering
University of South Florida
Tampa, Florida, USA
akurnikova@usf.edu

Fareeza Rahman

Department of Computer Science and
Engineering,
University of South Florida
Tampa, Florida, USA
fareeza@usf.edu

Julia Woodward

Department of Computer Science and
Engineering
University of South Florida
Tampa, Florida, USA
juliaevewoodward@usf.edu

Abstract

Augmented reality (AR) headsets are increasingly being used by people in a range of contexts. However, there are still open research questions on how people expect to interact with virtual objects in AR headsets, which limits our ability to design intuitive applications. We conducted an unconstrained elicitation study with 30 adults, in which participants showed how they would interact with a virtual cube in an AR headset. We did not constrain participants to a specific interaction technique (e.g., gestures vs. speech) and asked them open-ended questions to better understand their reasoning. For this paper, we focus on understanding the participants' expectations and mental models. Participants based their interactions on pop-culture, real-world experiences, and expected an immersive physically-engaging environment that follows the laws of physics. Our work contributes a deeper understanding of people's perceptions and expectations of interacting with AR headsets. We provide recommendations for designing future AR headset experiences.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; *Mixed / augmented reality*.

Keywords

human computer interaction, augmented reality, mental model, gesture elicitation, speech elicitation

ACM Reference Format:

Alejandro Delgado, Hetvi Shah, Md Mehedi Hasan Jibon, Anzhelika Kurnikova, Fareeza Rahman, and Julia Woodward. 2025. "Because we are human, we

use hands": Understanding People's Expectations and Mental Models of Virtual Object Interaction in AR Headsets. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719691>

1 Introduction

Augmented Reality (AR) is an emerging technology that can project virtual interactive elements onto the real world [45]. AR is being used in a variety of fields, including medicine [14, 50], education [23, 58], and manufacturing [3, 43]. Compared to other AR platforms (i.e., smartphones, tablets, computers), AR headsets are becoming a popular choice due to providing increased user immersion and hands-free capabilities [2, 34, 46, 47]. A key method of interaction in AR headsets is through gestures, which reflect the way people interact with the real world [39]. However, since gesture preference for interactions can vary across users [27], AR headsets often rely on a predefined set of arbitrary gestures that users must learn to navigate the interface [21]. Unfortunately, these gesture sets frequently fail to align with user expectations, resulting in non-intuitive interactions and an unpleasant user experience [21].

Recently, researchers have started conducting gesture elicitation studies with AR headsets (e.g., [38, 52]) to better understand how users interact with virtual objects and create more intuitive gesture sets. Although these studies have advanced our understanding of *how* users expect to interact with virtual objects in AR headsets, it remains unclear *why* users have these expectations and behaviors. Understanding the *why* behind users' expectations and behaviors is a key factor to determine their mental model, which contributes to the design of more intuitive gesture sets and applications [30]. A mental model can be defined as: "knowledge that forms the basis for users being able to construct reasonable actions; and explanations about why a set of actions is appropriate" [6].

Despite previous research looking into users' expectations for interacting with virtual objects in AR headsets [37, 38, 52], they constrained users to specific interactions (e.g., only gestures) and

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1395-8/25/04
<https://doi.org/10.1145/3706599.3719691>

did not focus on determining users' underlying mental models. Prior work has called for more research on users' mental models with AR headsets [21]. We expand on prior work by conducting an AR headset elicitation study with 30 adults to understand users' mental models of interacting with a virtual cube. Our approach involved (1) allowing participants complete freedom to choose their interaction technique (e.g., gestures, speech, etc.), and (2) asking open-ended questions about their interactions.

After completing the study sessions, we created an affinity diagram to analyze participants' responses and determine the main themes that constituted their underlying mental models. We found users relied on pop-culture, games, and real-world experiences for interacting with the virtual cube. Additionally, users expected an immersive physically-engaging AR environment that follows the laws of physics. The contributions of our work include (1) insights into users' mental models for interacting with a virtual object through an AR headset, and (2) a set of recommendations on designing future AR applications that match users' expectations and mental models. Our insights can inform the design and development of future AR applications. While this paper focuses on mental model analysis, we will complete quantitative analysis of participants' interaction preferences in future work.

2 Related Work

Gesture elicitation is a study design that aids in developing intuitive technologies [54]. In gesture elicitation studies, researchers gather input from users on how they expect to interact with different technologies. Participants are typically shown an *effect* (known as a referent) and are asked to perform the gesture they think would cause the effect. By analyzing different gestures taken from users, researchers can determine patterns and decide on gestures that would feel intuitive to a novice user. User-defined gestures are easier to learn, recall, and perform [54, 55]. Previous work has used gesture elicitation studies with different technologies such as tabletop computers [42, 55], mobile devices [16, 17, 29, 31, 40, 44], whole-body interfaces [11, 33], smart televisions [15], as well as speech recognition [7, 22]. Researchers looking to expand the way we interact with technology have also started to look into alternative gestures such as micro-gestures [8, 25], foot gestures [4, 18], nose gestures [36], ear gestures [9], and mid-air gestures [12, 49, 56]. Prior work has also started to conduct gesture elicitation studies with AR headsets to better understand users' expectations in an AR environment [4, 37, 38, 51–53, 57, 59].

Piumsomboon et al. [38] conducted an AR gesture elicitation study, in which participants would see an effect on a virtual toy block and then design the gesture for that effect. The authors found that the most commonly elicited gestures were those in which participants would interact with the virtual object as if it were a real object. Pham et al. [37] conducted an AR headset gesture elicitation study with three different-sized virtual objects: a miniature of a home, a small city model, and a model full-size room. They found that the size of the virtual object affected participants' behavior, such as using finger tips to pick up small objects. Williams et al. [52] conducted an AR headset gesture and speech elicitation study, in which participants saw a 3D cube through the headset. The cube executed different referents (e.g., moving, shrinking, rotating), and

participants were prompted to perform gesture, speech, and gesture plus speech interactions. The authors found that the participants mostly performed one-handed gestures that were easy to complete. These studies highlight the applicability of using elicitation to better understand users expectations; however, they explicitly prompted for each method of interaction (e.g., they ask a participant for a gesture, then speech, etc.) which fails to capture a user's first intuitive mode of interaction. To create AR systems that users find intuitive, it is crucial to capture their first impressions and behavior when interacting with a system. We expand on prior work by asking open-ended questions about their interactions rather than prompting for a specific mode of interaction.

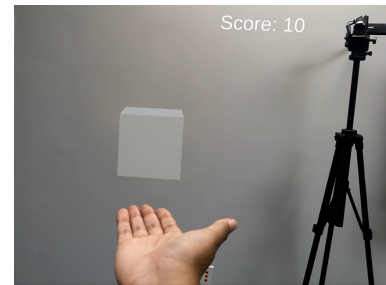


Figure 1: Participant's point-of-view from AR Headset.

3 Methodology

With our elicitation study, we explored how adults expect to interact with a virtual cube in an AR headset (Figure 1), as well as understanding their mental model behind their choices. We chose a cube as the virtual object because we wanted to compare our findings to previous work [51]. Choosing a cube also has the advantages of being an easily recognizable object, and referents such as rotation are easier to observe with a cube compared to using a sphere or cylinder [51]. For determining the referents used in our study we analyzed prior work [4, 11, 16, 29, 37, 51, 52, 55, 59], and selected the 17 most common referents (Table 1). We broke the list of 17 referents into three groups: (1) *Translation* which included referents that displaced the cube in space, (2) *Manipulation* in which the cube's properties were modified, and (3) *Abstract* in which referents were not commonly associated with the previous two categories. We referenced Williams et al. [51] for the creation of the referent groups. Each group contained about 6 cube referents (e.g., *Move Up*, *Move Down*), and the order in which the groups were shown was counterbalanced across participants. The order of the referents inside of the groups remained constant. The virtual cube was placed in front of the participant's eye line according to their height. Participants initially stood on an X outlined on the floor while they were shown a referent such as *Zoom In* (i.e., cube gets bigger and closer) to make sure participants always viewed the referent from the same angle and position. After the referent completed, participants were informed they could move around the room to show how they would complete their interaction. Participants were able to take breaks at any point during the study. The study was conducted in a spacious room that allowed for participants' unconstrained movement and it took approximately 60

minutes. Our research protocol was approved by our Institutional Review Board, and participants were compensated \$40.

Table 1: Referents

Group	Referents
Translation	Move (Up/Down) Move (Left/Right) Move (Towards/Away)
Manipulation	Make (Bigger/Smaller) Rotate (X/Y/Z) axis Zoom (In/Out)
Abstract	Create Destroy Select (One/Multiple)

Our study was a within-subject study (i.e., participants saw all referents). Participants first completed a short demographic survey about their age, gender, and prior experience with VR and AR headsets. After participants finished the survey, they then put on the AR headset and completed a short practice, in which they saw the virtual cube in the headset and could walk around the room. After participants finished the practice section, the main part of the study began in which participants would see a referent, such as the cube moving up, and we would then ask them: "How would you make the cube [current referent, e.g. Move Up] like you saw?" Participants then showed us how they would complete the interaction (e.g., physically acting out the interaction, giving a verbal command). We informed participants beforehand: "You can use any part of your body or speak to it [the cube]" so they were aware they were not constrained to a specific mode of interaction. We made the question open-ended to capture participants' first choice for a mode of interaction with the cube. After they answered the first question we asked: "Why did you want to [Move Up] the cube like that?" We then asked participants "How would you make the cube [Move Up] in another way?" and repeat the same procedure as we did for their first interaction, in order to get their top two choices.

For the AR headset we opted for a *Magic Leap 2* [24] due to its optimized weight distribution, allowing for a more comfortable experience for extended use. The headset has a FoV of 45H x 55V (70D) and a 1440 x 1760 resolution per eye. To display the referents, we created an application using *Unity* version 2022.3.22f1 and the *Magic Leap 2 SDK* [24]. We included a scoreboard for gamification purposes. The cube was placed approximately 60 centimeters (cm) away from the headset with a size of 10 cm³. The movement of the referents remained in the headset's FoV, reducing the need for participants to pan around the environment to observe the referent. To view the referent animations and development details, see Supplementary Material.

3.1 Participants and Data Analysis

The study involved 30 adults between the ages of 18 and 54 (M=31.4, SD=12), 13 females and 17 males. Out of the 30 participants, 27 were right-handed and 18 self-reported previous experience with VR headsets. Only one participant self-reported previous use with

an AR headset through participation in another research study. The data collected from our study included audio and video recordings of the participants, and recordings of the participants' point-of-view from the AR headset. We analyzed the participants' explanations through affinity diagramming, which is a method to categorize large scale qualitative data through a bottom-up inductive approach [5]. We did not compute inter-rater reliability as it is not recommended when the research goal is to determine concepts and themes [26]. To create the affinity diagram, we iteratively grouped 1,013 participant explanations (30 participants x 17 referents x 2 interactions per referent) into themes using *Lucidchart* [1], an online whiteboard tool for remote collaboration; seven explanations were omitted due to incomplete or missing data. The affinity diagram allowed us to identify recurring themes and patterns regarding the participants' expectations and preferences for interacting with virtual objects in AR headsets. We categorized the qualitative responses we received from participants into 7 overarching themes described below.

4 Findings

4.1 User's Prior Experience

We found that participants utilized their prior experience with *Existing Technology*, *Touchscreen* interfaces, *Games*, *Pop Culture*, and the *Real World* to construct their mental models for interacting with the virtual cube. Participants relied on their interactions with *Existing Technology* for creating new cube interactions: "Because that's how we zoom in on phones. Very easy to learn, intuitive and probably you won't even have to teach people." (P18). Additionally, participants utilized their knowledge of *Pop Culture*, *Games*, and the *Real World* to aid with their interaction propositions. P5 mentioned "Cause I am a Harry Potter fan", when explaining why they used one finger to rotate the cube in a wand-like motion. For *Destroy*, P14 chose a snapping gesture, "Snapping means clear, I just thought of Thanos", inspired by the iconic "snap" from the movie *Avengers: Infinity War* [41]. In a more playful real-world example, for *Rotate*, P5 used a one-finger gesture: "Cause that's how I tell my dog to roll over." This shows how participants used familiarity to guide how they would manipulate objects in AR – if it works in real life, they figured it would work in the virtual environment as well.

4.2 Users' Interaction Expectations

We found that participants considered the level of *System Awareness* when creating their interactions: "A cube is a general thing, the software should know what the cube is...", with some participants proposing using their mind for abstract referents such as *Create*: "The mind is the fastest, the nerves are the fastest right, it would take time for the hand gestures to do that, but thinking is the fastest way." (P13). Overall, participants expected the use of *Hands* or *Direct Manipulation* to work on the cube: "Well, if it was a real object I would try to move it with my hand. Because if I can physically touch it then I can manipulate it." (P29). Additionally, participants preferred *Inverse* or *Mirrored* interactions for dichotomous referents such as *Zoom In/Out*. Another expectation by participants was to have a *Sense of Control* to ensure precise interactions. This preference led some participants towards interactions they considered to have better visual feedback: "...Because that way I can almost be precise with my measurements and get it to a size that I want visibly." (P9).

Other participants felt they had more control over their interactions by using speech: *"I prefer to have keywords so that I know that they would be precise."* (P29). Furthermore, participants thought of using an *External Object* to aid with their interactions, such as mentioning the use of a hammer for the *Destroy* referent. Participants also considered external objects with clear affordances (e.g., handle for turning) to aid in their interactions: *"Could pretend there is a handle and rotate in directions that you want..."* (P3). This idea of using external objects provides a peak into participants' mental models, in which they take an action they would perform on a real object and apply it to the virtual cube. Norman et al. [32] explains how affordances help people intuitively interact with objects without the need for instructions. We talk more about using external objects to improve gesture sets in our Discussion section.

4.3 Human Characteristics

We observed that some participants explicitly mentioned *Human Characteristics* as an inspiration for their interactions. They described their interactions in terms of what entails to be *Human* and a sense of *Ownership* over the virtual cube. For instance, P2 explained, *"Basically, I think people want to use it like it's a real object, I think that's the human action."* P16 expressed their reliance on habits, *"It is just from habit...easier it makes sense in your brain, just habits"*, and P2 simply expressed, *"Because we are human, we use hands."* Participants also leaned into interactions that fostered a feeling of *Ownership* over the cube: *"That motion of it coming closer, is like it is mine, I made it. I feel like I have more control of the object."* (P1). Previous research has shown that the simple act of touching an object increases the sense of perceived ownership [35]. Because these participants were touching the virtual cube as if it was a real object, they had an increased sense of ownership and confidence with their interactions.

4.4 Characteristics of Users' Interactions

We found that participants expected interactions that were *Quick*, *Simple*, and *Easy* to do. We also observed a preference for interactions that require minimal *Effort*, feel *Intuitive/Natural*, and provide a feeling of a *Fluid/Smooth Movement*. Interestingly, we had participants explicitly express a preference for interacting with the cube, such as wanting to only use gestures. We also observed that participants often had a mixed use approach by switching between interaction modes depending on the referent (e.g., using speech for *Create* and then a gesture for *Destroy*). Participants also preferred interactions they perceived required minimal effort; P2 encapsulated this preference: *"Sometimes speech is easier. We are lazy, like Alexa find something. We don't want to press the button, we want to talk."* Participants also preferred for an interaction to feel *Intuitive/Natural*: *"It just feels more natural, if I'm in a natural environment and those are some real objects that I have in front of me..."* (P13). Additionally, participants leaned towards gestures that felt were *Fluid* and had a *Smooth Movement*: *"It seems very fluid to rotate it do it that way..."* (P6). Furthermore, participants wanted their interactions to be *Interactive*, *Physically Engaging*, and *Fun* to do. Participants expressed how gestures, which manipulated the cube as if it was a real physical object, felt more *Interactive*: *"...it feels like I am more interacting with the object if I just do it with*

my hands." (P17). We also found that participants often sought to perform interactions in ways they found *Physically Engaging* and *Fun*: *"...a more fun way would be to bop it. It requires more activity, so it may be more fun to do."* (P3). The tendency towards physically active interaction seems to align with observations made by Gerling et al. [19], in which they found that full-body motion-control games can enhance users' mood and emotional well-being.

4.5 Utilization of Physics

Throughout their interactions, participants intuitively relied on the *Utilization of Physics* for their interactions. This common sense knowledge about the physical world, also known as *naive physics* [20], was applied by participants when interacting with the properties of the cube, such as its *Size*, perceived *Weight*, as well as external influences such as physical *Force* and *Air*. The *Size* and possible *Weight* of the cube influenced how the participants thought of their interactions. As P22 stated, *"If the cube is small enough I can grab with my hand and pull it down..."* P5 wanted to push the cube with only one finger: *"Looks light enough to be able to do that"*, while P10 was weary of how heavy the cube could be: *"...But this thing (cube) will strain my hand if the weight is heavy"*. This indicates that participants will interact with a virtual object differently depending on its *Size* and *Weight*, similar to how they would interact with a real-world object. Our observations appear to align with what was observed by Pham et al. [37], in which participants modified the size of their gestures according to the virtual object's size. Participants also factored in the use of *Force* in their cube interactions: *"Maybe like just flicking it up ... I'd have to be careful depending on the speed like how far I wanted it to go."* (P23). Additionally, participants utilized *Air* as a way to manipulate the cube, such as physically blowing air with their mouth: *"I would blow on it...So that it could spin with speed."* (P9). Participants also experimented with creating an air current under the cube: *"you could move your hand underneath it and air it to move up."* (P11). We speculate that this idea could have been based on their interpretation of *naive physics*, which correlated the floating virtual cube with real-life floating objects that could be affected by the force of air.

4.6 Users' Perception

A major contributor towards participants' mental models was their *Perception*. Participants' *Environment*, sense of *Immersion*, and *Spatial* perception shaped how they interacted with the cube. Some participants created their mental models by visualizing the cube as sitting on top of a surface, possibly as an attempt to align more with real-world experiences: *"Imagine the cube is on the flat surface, you kind of push it to move it in the direction you want it to go"* (P8), while other participants used the cube's placement in mid-air to create unique interactions such as: *"In this case it's [the cube] not related with a physical object because there wouldn't be gravity. So in this case, it's just like a floating object so then I would put it down like it was floating."* (P27). Participants expressed how using gestures provided an increased sense of *Immersion*: *"Because it's more interactive and yes, like...you feel more immersed. You can touch the object and you feel like you have an influence on the world instead of just saying something and having it happen."* (P18). Another factor which influenced participants' mental models was their *Spatial* perception.

We found that participants often conceptualized their interaction in 2D and then translated it to 3D space. As an example, for *Create*, P9 explained, "Well first I'm creating the cube from the 2D shape of the square. So I'm using the 2D shape of the square and projecting it to make a 3D cube." P18 used similar thinking for *Make Smaller*, "Because, that's how you would usually resize a 2D object. So, I guess it's easier to remember if that's translated to 3D." Interestingly, we observed some participants move themselves instead of moving the cube: "I'm ...moving forward towards the object. I think it is similar in the real world" (P7). This could be due to participants replicating how they would act with a real-world object.

4.7 User Limitations

Participants considered potential challenges that they or other users might face while interacting with the virtual cube. These limitations included difficulties such as *Not Able To Use Hands*, varying levels of *User Ability*, managing *Body Limitations*, and handling *Multitasking*. Some participants considered possible situations that they were *Not Able To Use Hands* for their interactions, possibly due to physical or situational limitations: "...saying 'rotate to the left'... In case if you might not have physical means to rotate the object." (P6). P18 also chose speech, noting, "It's an easy alternative [to] just grabbing it, if your hands are full." Some participants expanded on this idea, emphasizing the utility of speech interactions for *Multitasking*: "...with the voice I can work anything else and also I can do this job too." (P22). Participants constrained some of their interactions based on their self-perceived *User Ability*: "I am not a good drawer so I wouldn't draw it..." (P1). Some participants even considered the abilities of hypothetical users: "Easier, especially if you are say Quadriplegic and you don't have any physical use and only thing you have is voice control." (P16). *Body Limitations*, including physical discomfort, were mentioned by participants in their interactions: "But there's a question, we try to move it far away, but there's a limit of our body. So how far we want to move it I don't know" (P2), and "It seems like it would be easier to put the pressure on it that way and not have to hurt my hands." (P11). We believe these are determining factors used by participants as part constructing their mental model for gestures that are within physical limitations.

5 Discussion

5.1 Summary of Our Findings vs. Prior Work

In comparing our findings with previous work, we found overlapping observations, such as the properties of a virtual object (e.g., size, weight) affecting how users interact with it, and the prevalence of legacy bias [12, 37, 38, 55]. Legacy bias can be understood as the bias that users have from using previous technologies and applications [28]. Our participants wanted to choose interactions that felt familiar and intuitive, which were often gestures or speech interactions borrowed from existing technologies. Interestingly, participants also borrowed the terminology used with other technologies, such as "swipe", "click", and "scroll" to describe their interactions with the virtual cube. Regarding the size of a virtual object, Pham et al. [37] found that participants would elicit different gestures depending on the size of the object. Prior research has shown that people prefer *physical* gestures for virtual object interactions [38, 55], in which they act physically on the virtual object as if it was a real

object (e.g., grabbing the virtual object and relocating it for a move referent). We also observed a preference for physical gestures, with an emphasis on the use of hands. Participants generally perceived hand gestures as easier and more intuitive for interacting with the cube. Additionally, the participants in our study expressed how the use of their hands made their interactions more immersive, engaging, and fun.

As we mentioned in *Utilization of Physics*, placing the cube as a floating object in the AR environment influenced participants' interactions. For example, some participants' mental model assimilated the floating cube with floating objects in real life: "...if there was something floating in the air like a cloud or something - a cloud of steam, that's how I'd put it up." (P23). We believe this assimilation led to some participants using air as way to manipulate the cube since real-world floating objects can be affected by air (e.g., balloons, clouds). Additionally, participants used the fact that the cube was a floating object to interact with it in unique ways, such as P27 creating interactions based on their assumption that there would not be any gravity since the cube was floating. Our findings on legacy bias, the influence of the cube's properties and placement, and why physical gestures were preferred add depth to the understanding of users' expectations and mental models of virtual-object interaction.

5.2 Design Recommendations

Our findings revealed that participants tend to have a preferred way of interacting with the cube (e.g., mainly using their hands). However, as we discussed in *Users' Interaction Expectations*, when participants wanted more precise interactions, some leaned more towards the use of speech: "With the voice you just say, oh rotate 45 degrees. If you are not happy with this, okay 40... But with the pop-up thing [referring to doing a gesture] you would have to like keep changing like that" (P18). Additionally, as mentioned in *Characteristics of Users' Interactions*, participants switched between modes of interaction depending on the referent. As an example, for *Rotate*, P22 did a gesture, "...if want to rotate something I always prefer my hand to do that", while for *Create*, "Oh I prefer voice command here. That's the easiest way." Korkiakoski et al. [22] studied the preferences for pre-defined voice commands and gestures in AR applications. They found that initially participants felt most comfortable with voice commands. However, when researchers explicitly asked participants for their preference in a real-world scenario, participants opted for a mixed-use of both modes. We believe that the mixed-use of interaction modes will better account for the individual preferences of users and lead to more intuitive interactions. **Given this insight, future AR applications should provide a seamless transition between the use of gestures, voice commands, and other modes of interaction.**

In our study, we found that participants often made use of analogies of their lived experience (e.g. real world, games, pop culture) to construct their mental models for interacting with the cube. Previous research has looked into how people construct mental models [10] and found that analogies are an effective way for people to understand new concepts. **We believe that the implementation of analogies as teaching aids could help users construct a valid mental model for more intuitive interactions with AR applications.** Khurana et al. [21] investigated how novice users

discover gestural interactions with consumer AR applications. They found that a majority of participants lacked a mental model about how to use gestures to interact with an AR application. In *Users' Interaction Expectations*, we found that some participants imagined using external objects with clear affordances to aid with gesture creation. These objects helped the participant connect their experience with a familiar action to create a new, yet intuitive interaction. These kinds of visual aids could expand the user's mental model by connecting a real-world action with the corresponding application specific gesture, leading to a more intuitive understanding of the gesture and how to use it. While existing AR applications do provide some visual feedback, prior work has found that it is unclear for users [21]. **We recommend the implementation of external objects as visual aids (e.g., virtual handle animation for rotating) as they could improve overall usability and intuitiveness in AR applications.** In addition, our findings demonstrated that participants often relied on their understanding of *naive physics* for their interactions with the cube. Participants expected the cube to be affected just as a real-world object would, which aligns with prior findings [38, 51, 55]. **Therefore, we advise the incorporation of a physics engine to enhance virtual-object interaction.**

6 Conclusion, Limitations, and Future Work

Although AR headsets are increasingly being used, there are still open research questions on how people expect to interact with virtual objects in the headsets. Therefore, we conducted an unconstrained AR headset elicitation study with 30 adults. While 30 might seem small, it is consistent with prior work [13, 33, 36, 48, 51]. We found that participants base their interactions on pop-culture and real-world experiences and want multi-modal interactions. In addition, we recommend using analogies to inform users about how to interact with virtual objects in AR headsets. It is important to mention, we used a virtual cube, which might have influenced participants' interaction choices. Future research could explore interactions in more diverse and tailored AR scenarios for specific contexts (e.g., anchored objects, productivity, gaming). Our study did not examine how factors such as field of view constraints, occlusion, or depth perception differences could influence users' expectations. Future work could analyze if these factors significantly influence users' expectations in AR headsets. Also, our chosen referent animations may have influenced participants' interactions (see Supplementary Material for all referent animations). For example, the *Destroy* referent animation depicted the cube breaking into pieces, which led some participants to try to mimic the destruction effect in their interactions. Future work could explore alternative ways to present referents to reduce influence from animation effects. Furthermore, while this paper focuses on participants' explanations, for future work, we will perform a quantitative analysis on participants' gesture and speech preferences from this study. Another potential direction for future research is to explore how different populations, such as children, think about and interact with virtual objects in an AR environment.

Acknowledgments

We thank our participants for helping us with our study.

References

- [1] Lucidchart. 2008. Lucidchart |Diagramming Powered By Intelligence. <https://www.lucidchart.com/pages/>.
- [2] Thomas Alsop. 2024. Topic: AR Glasses and Headsets. <https://www.statista.com/topics/10134/ar-glasses/>.
- [3] George Apostolopoulos, Dionisis Andronas, Nikos Fourtakas, and Sotiris Makris. 2022. Operator Training Framework for Hybrid Environments: An Augmented Reality Module Using Machine Learning Object Recognition. *Procedia CIRP* 106 (Jan. 2022), 102–107. <https://doi.org/10.1016/j.procir.2022.02.162>
- [4] Christopher R. Austin, Barrett Ens, Kadek Ananta Satriadi, and Bernhard Jenny. 2020. Elicitation Study Investigating Hand and Foot Gesture Interaction for Immersive Maps in Augmented Reality. *Cartography and Geographic Information Science* 47, 3 (May 2020), 214–228. <https://doi.org/10.1080/15230406.2019.1696232>
- [5] Hugh Beyer and Karen Holtzblatt. 1999. Contextual Design. *Interactions* 6, 1 (Jan. 1999), 32–42. <https://doi.org/10.1145/291224.291229>
- [6] John M. Carroll and Judith Reitman Olson. 1988. Mental Models in Human-Computer Interaction. In *Handbook of Human-Computer Interaction*, MARTIN Helander (Ed.), North-Holland, Amsterdam, 45–65. <https://doi.org/10.1016/B978-0-444-70536-5.50007-5>
- [7] Nabil Al Nahin Ch, Diana Tosca, Tyanna Crump, Alberta Ansah, Andrew Kun, and Orit Shaer. 2022. Gesture and Voice Commands to Interact With AR Windshield Display in Automated Vehicle: A Remote Elicitation Study. In *Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '22)*. Association for Computing Machinery, New York, NY, USA, 171–182. <https://doi.org/10.1145/3543174.3545257>
- [8] Edwin Chan, Teddy Seyed, Wolfgang Stuerzlinger, Xing-Dong Yang, and Frank Maurer. 2016. User Elicitation on Single-hand Microgestures. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. Association for Computing Machinery, New York, NY, USA, 3403–3414. <https://doi.org/10.1145/2858036.2858589>
- [9] Yu-Chun Chen, Chia-Ying Liao, Shuo-wen Hsu, Da-Yuan Huang, and Bing-Yu Chen. 2020. Exploring User Defined Gestures for Ear-Based Interactions. *Proceedings of the ACM on Human-Computer Interaction* 4, ISS (Nov. 2020), 186:1–186:20. <https://doi.org/10.1145/3427314>
- [10] Allan Collins and Dedre Gentner. 1987. How People Construct Mental Models. In *Cultural Models in Language and Thought* (1 ed.), Dorothy Holland and Naomi Quinn (Eds.), Cambridge University Press, Cambridge, 243–266. <https://doi.org/10.1017/CBO9780511607660.011>
- [11] Sabrina Connell, Pei-Yi Kuo, Liu Liu, and Anne Marie Piper. 2013. A Wizard-of-Oz Elicitation Study Examining Child-Defined Gestures with a Whole-Body Interface. In *Proceedings of the 12th International Conference on Interaction Design and Children (IDC '13)*. Association for Computing Machinery, New York, NY, USA, 277–280. <https://doi.org/10.1145/2485760.2485823>
- [12] Jian Cui, Arjan Kuijper, Dieter W. Fellner, and Alexei Sourin. 2016. Understanding People's Mental Models of Mid-Air Interaction for Virtual Assembly and Shape Modeling. In *Proceedings of the 29th International Conference on Computer Animation and Social Agents (2016-05-23) (CASA '16)*. Association for Computing Machinery, New York, 139–146. <https://doi.org/10.1145/2915926.2919330>
- [13] Andreea Danieleescu and David Piorkowski. 2022. Iterative Design of Gestures During Elicitation: Understanding the Role of Increased Production. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3491102.3501962>
- [14] Lucio Tommaso De Paolis, Sergio Teodoro Vite, Miguel Ángel Padilla Castañeda, César Fabian Domínguez Velasco, Salvatore Muscatello, and Aldo Francisco Hernández Valencia. 2022. An Augmented Reality Platform with Hand Gestures-Based Navigation for Applications in Image-Guided Surgery: Prospective Concept Evaluation by Surgeons. *International Journal of Human-Computer Interaction* 38, 2 (Jan. 2022), 131–143. <https://doi.org/10.1080/10447318.2021.1926116>
- [15] Haiwei Dong, Ali Danesh, Nadia Figueroa, and Abdulmoteleb El Saddik. 2015. An Elicitation Study on Gesture Preferences and Memorability Toward a Practical Hand-Gesture Vocabulary for Smart Televisions. *IEEE Access* 3 (2015), 543–555. <https://doi.org/10.1109/ACCESS.2015.2432679>
- [16] Ze Dong, Thammathip Piumsomboon, Jingjing Zhang, Adrian Clark, Huidong Bai, and Rob Lindeman. 2020. A Comparison of Surface and Motion User-Defined Gestures for Mobile Augmented Reality. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3382883>
- [17] Shariff A. M. Faleel, Michael Gammon, Yumiko Sakamoto, Carlo Menon, and Pourang Irani. 2020. User Gesture Elicitation of Common Smartphone Tasks for Hand Proximate User Interfaces. In *Proceedings of the 11th Augmented Human International Conference (AH '20)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3396339.3396363>
- [18] Yasmin Felberbaum and Joel Lanir. 2018. Better Understanding of Foot Gestures: An Elicitation Study. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New

- York, NY, USA, 1–12. <https://doi.org/10.1145/3173574.3173908>
- [19] Kathrin Maria Gerling. 2014. Motion-based game interaction for older adults. *ACM SIGACCESS Accessibility and Computing* 108 (Jan. 2014), 27–30. <https://doi.org/10.1145/2591357.2591361>
- [20] Robert J.K. Jacob, Audrey Girouard, Leanne M. Hirshfield, Michael S. Horn, Orit Shaer, Erin Treacy Solovey, and Jamie Zigelbaum. 2008. Reality-based interaction: a framework for post-WIMP interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08)*. Association for Computing Machinery, New York, NY, USA, 201–210. <https://doi.org/10.1145/1357054.1357089>
- [21] Anjali Khurana, Michael Glueck, and Parmit K. Chilana. 2024. Do I Just Tap My Headset? How Novice Users Discover Gestural Interactions with Consumer Augmented Reality Applications. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4 (Jan. 2024), 165:1–165:28. <https://doi.org/10.1145/3631451>
- [22] Mikko Korkiakoski, Paula Alavesä, and Panos Kostakos. 2024. Preference in Voice Commands and Gesture Controls With Hands-Free Augmented Reality With Novel Users. *IEEE Pervasive Computing* 23, 1 (Jan. 2024), 18–26. <https://doi.org/10.1109/MPRV.2024.3364541>
- [23] Hong-Quan Le and Jee-In Kim. 2017. An Augmented Reality Application with Hand Gestures for Learning 3D Geometry. In *2017 IEEE International Conference on Big Data and Smart Computing (BigComp)*. IEEE, Korea, 34–41. <https://doi.org/10.1109/BIGCOMP.2017.7881712>
- [24] Magic Leap. 2022. Magic Leap 2: Next-Gen Augmented Reality Headset for Professionals. <https://www.magicleap.com/magic-leap-2>
- [25] Nathan Magrofuoco, Jorge-Luis Pérez-Medina, Paolo Roselli, Jean Vanderdonck, and Santiago Villarreal. 2019. Eliciting Contact-Based and Contactless Gestures With Radar-Based Sensors. *IEEE Access* 7 (2019), 176982–176997. <https://doi.org/10.1109/ACCESS.2019.2951349>
- [26] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and Inter-rater Reliability in Qualitative Research: Norms and Guidelines for CSCW and HCI Practice. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (Nov. 2019), 72:1–72:23. <https://doi.org/10.1145/3359174>
- [27] David McNeill. 2005. *Gesture and Thought*. University of Chicago Press, Chicago, IL, US, xii, 318 pages. <https://doi.org/10.7208/chicago/9780226514642.001.0001>
- [28] Meredith Ringel Morris, Jacob O. Wobbrock, and Andrew D. Wilson. 2010. Understanding users' preferences for surface gestures. In *Proceedings of Graphics Interface 2010* (Ottawa, Ontario, Canada) (GI '10). Canadian Information Processing Society, CAN, 261–268. <https://doi.org/10.5555/1839214.1839260>
- [29] Vijayakumar Nanjappan, Rongkai Shi, Hai-Ning Liang, Haoru Xiao, Kim King-Tong Lau, and Khalad Hasan. 2019. Design of Interactions for Handheld Augmented Reality Devices Using Wearable Smart Textiles: Findings from a User Elicitation Study. *Applied Sciences* 9, 15 (Jan. 2019), 3177. <https://doi.org/10.3390/app9153177>
- [30] Anja Naumann, Jörn Hurtienne, Johann Habakuk Israel, Carsten Mohs, Martin Christof Kindsmüller, Herbert A. Meyer, and Steffi Huflein. 2007. Intuitive Use of User Interfaces: Defining a Vague Concept. In *Engineering Psychology and Cognitive Ergonomics*, Don Harris (Ed.). Springer, Berlin, Heidelberg, 128–136. https://doi.org/10.1007/978-3-540-73331-7_14
- [31] Chloe Ng and Nicolai Marquardt. 2022. Eliciting User-Defined Touch and Mid-air Gestures for Co-located Mobile Gaming. *Proceedings of the ACM on Human-Computer Interaction* 6, ISS (Nov. 2022), 569:303–569:327. <https://doi.org/10.1145/3567722>
- [32] Don Norman. 2013. *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books, New York.
- [33] Francisco R. Ortega, Katherine Tarre, Mathew Kress, Adam S. Williams, Armando B. Barreto, and Naptali D. Rishé. 2019. Selection and Manipulation Whole-Body Gesture Elicitation Study In Virtual Reality. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, Japan, 1723–1728. <https://doi.org/10.1109/VR.2019.8798105>
- [34] Ahmed Oun, Nathan Hagerdorn, Caleb Scheidegger, and Xiangyi Cheng. 2024. Mobile Devices or Head-Mounted Displays: A Comparative Review and Analysis of Augmented Reality in Healthcare. *IEEE Access* 12 (2024), 21825–21839. <https://doi.org/10.1109/ACCESS.2024.3361833>
- [35] Joann Peck and Suzanne B. Shu. 2009. The Effect of Mere Touch on Perceived Ownership. *Journal of Consumer Research* 36, 3 (Oct. 2009), 434–447. <https://doi.org/10.1086/598614>
- [36] Jorge-Luis Pérez-Medina, Santiago Villarreal, and Jean Vanderdonck. 2020. A Gesture Elicitation Study of Nose-Based Gestures. *Sensors* 20, 24 (Jan. 2020), 7118. <https://doi.org/10.3390/s20247118>
- [37] Tran Pham, Jo Vermeulen, Anthony Tang, and Lindsay MacDonald Vermeulen. 2018. Scale Impacts Elicited Gestures for Manipulating Holograms: Implications for AR Gesture Design. In *Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18)*. Association for Computing Machinery, New York, NY, USA, 227–240. <https://doi.org/10.1145/3196709.3196719>
- [38] Thammathip Piumsomboon, Adrian Clark, Mark Billingham, and Andy Cockburn. 2013. User-Defined Gestures for Augmented Reality. In *Human-Computer Interaction – INTERACT 2013*, Paula Kotzé, Gary Marsden, Gitte Lindgaard, Janet Wesson, and Marco Winckler (Eds.). Springer, Berlin, Heidelberg, 282–299. https://doi.org/10.1007/978-3-642-40480-1_18
- [39] Kasper Rise and Ole Andreas Alsos. 2020. The Potential of Gesture-Based Interaction. In *Human-Computer Interaction. Multimodal and Natural Interaction*, Masaaki Kurosu (Ed.). Springer International Publishing, Cham, 125–136. https://doi.org/10.1007/978-3-030-49062-1_8
- [40] Jaime Ruiz, Yang Li, and Edward Lank. 2011. User-Defined Motion Gestures for Mobile Interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*. Association for Computing Machinery, New York, NY, USA, 197–206. <https://doi.org/10.1145/1978942.1978971>
- [41] Anthony Russo and Joe Russo. 2018. *Avengers: Infinity War*.
- [42] Karen Rust, Meethu Malu, Lisa Anthony, and Leah Findlater. 2014. Understanding Childdefined Gestures and Children's Mental Models for Touchscreen Tabletop Interaction. In *Proceedings of the 2014 Conference on Interaction Design and Children (IDC '14)*. Association for Computing Machinery, New York, NY, USA, 201–204. <https://doi.org/10.1145/2593968.2610452>
- [43] Frerk Saxen, Anne Köpsel, Simon Adler, Rüdiger Mecke, Ayoub Al-Hamadi, Johannes Tümler, and Anke Huckauf. 2017. Investigation of an Augmented Reality-Based Machine Operator Assistance-System. In *Companion Technology: A Paradigm Shift in Human-Technology Interaction*, Susanne Biundo and Andreas Wendemuth (Eds.). Springer International Publishing, Cham, 471–483. https://doi.org/10.1007/978-3-319-43665-4_23
- [44] Shaikh Shawon Arefin Shimon, Sarah Morrison-Smith, Noah John, Ghazal Fahimi, and Jaime Ruiz. 2015. Exploring User-Defined Back-Of-Device Gestures for Mobile Devices. In *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '15)*. Association for Computing Machinery, New York, NY, USA, 227–232. <https://doi.org/10.1145/2785830.2785890>
- [45] D.W.F. Van Krevelen and R. Poelman. 2010. A Survey of Augmented Reality Technologies, Applications and Limitations. *International Journal of Virtual Reality* 9, 2 (Jan. 2010), 1–20. <https://doi.org/10.20870/IJVR.2010.9.2.2767>
- [46] Sergei Vardomatski. 2021. Council Post: Augmented And Virtual Reality After Covid-19. <https://www.forbes.com/councils/forbestechcouncil/2021/09/14/augmented-and-virtual-reality-after-covid-19/>
- [47] Kenneth J. Varnum. 2019. *Beyond Reality: Augmented, Virtual, and Mixed Reality in the Library*. American Library Association, Chicago.
- [48] Santiago Villarreal-Narvaez, Jean Vanderdonck, Radu-Daniel Vatavu, and Jacob O. Wobbrock. 2020. A Systematic Review of Gesture Elicitation Studies: What Can We Learn from 216 Studies?. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 855–872. <https://doi.org/10.1145/3357236.3395511>
- [49] Panagiotis Vogiatzidakis and Panayiotis Koutsabasis. 2020. Mid-Air Gesture Control of Multiple Home Devices in Spatial Augmented Reality Prototype. *Multimodal Technologies and Interaction* 4, 3 (Sept. 2020), 61. <https://doi.org/10.3390/mti4030061>
- [50] Shiyao Wang, Michael Parsons, Jordan Stone-McLean, Peter Rogers, Sarah Boyd, Kristopher Hoover, Oscar Meruvia-Pastor, Minglun Gong, and Andrew Smith. 2017. Augmented Reality as a Telemedicine Platform for Remote Procedural Training. *Sensors* 17, 10 (Oct. 2017), 2294. <https://doi.org/10.3390/s17102294>
- [51] Adam S. Williams, Jason Garcia, and Francisco Ortega. 2020. Understanding Multimodal User Gesture and Speech Behavior for Object Manipulation in Augmented Reality Using Elicitation. *IEEE Transactions on Visualization and Computer Graphics* 26, 12 (Dec. 2020), 3479–3489. <https://doi.org/10.1109/TVCG.2020.3023566>
- [52] Adam S. Williams and Francisco R. Ortega. 2020. Understanding Gesture and Speech Multimodal Interactions for Manipulation Tasks in Augmented Reality Using Unconstrained Elicitation. *Proceedings of the ACM on Human-Computer Interaction* 4, ISS (Nov. 2020), 1–21. <https://doi.org/10.1145/3427330>
- [53] Adam S. Williams and Francisco R. Ortega. 2022. The Impacts of Referent Display on Gesture and Speech Elicitation. *IEEE Transactions on Visualization and Computer Graphics* 28, 11 (Nov. 2022), 3885–3895. <https://doi.org/10.1109/TVCG.2022.3203090>
- [54] Jacob O. Wobbrock, Htet Htet Aung, Brandon Rothrock, and Brad A. Myers. 2005. Maximizing the Guessability of Symbolic Input. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems (CHI EA '05)*. Association for Computing Machinery, New York, NY, USA, 1869–1872. <https://doi.org/10.1145/1056808.1057043>
- [55] Jacob O. Wobbrock, Meredith Ringel Morris, and Andrew D. Wilson. 2009. User-Defined Gestures for Surface Computing. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '09)*. Association for Computing Machinery, New York, NY, USA, 1083–1092. <https://doi.org/10.1145/1518701.1518866>
- [56] Yiqi Xiao, Ke Miao, and Chenhan Jiang. 2021. Mapping Directional Mid-Air Unistroke Gestures to Interaction Commands: A User Elicitation and Evaluation Study. *Symmetry* 13, 10 (Oct. 2021), 1926. <https://doi.org/10.3390/sym13101926>
- [57] Yukang Yan, Xin Yi, Chun Yu, and Yuanchun Shi. 2019. Gesture-Based Target Acquisition in Virtual and Augmented Reality. *Virtual Reality & Intelligent Hardware* 1, 3 (June 2019), 276–289. <https://doi.org/10.3724/SP.J.2096-5796.2019>

0007

- [58] Mau-Tsuen Yang and Wan-Che Liao. 2014. Computer-Assisted Culture Learning in an Online Augmented Reality Environment Based on Free-Hand Gesture Interaction. *IEEE Transactions on Learning Technologies* 7, 2 (April 2014), 107–117. <https://doi.org/10.1109/TLT.2014.2307297>
- [59] Xiaoyan Zhou, Adam Sinclair Williams, and Francisco Raul Ortega. 2022. Eliciting Multimodal Gesture+Speech Interactions in a Multi-Object Augmented Reality Environment. In *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology (VRST '22)*. Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3562939.3565637>